

2026年4月25日 物理 CSCA考试

April 2026 CSCA Physics Exam

1. 下列物理量中属于标量的是 ()

Which of the following physical quantities is a **scalar**? ()

- A. 位移 Displacement
- B. 加速度 Acceleration
- C. 力 Force
- D. 功 Work

2. 一个物体在月球上的重力约为地球上的 $1/6$ 。若物体质量 $m = 6 \text{ kg}$ ，地球上的重力加速度 $g = 10 \text{ m/s}^2$ ，则它在月球上的重力为 ()

An object's weight on the Moon is about $1/6$ of its weight on Earth. If the object has a mass $m = 6 \text{ kg}$ and the acceleration due to gravity on Earth $g = 10 \text{ m/s}^2$, its weight on the Moon is ().

- A. 36 N
- B. 10 N
- C. 60 N
- D. 6 N

3. 一根弹簧的劲度系数 $k = 50 \text{ N/m}$ ，原长 10 cm 。若将其拉长至 14 cm ，拉力大小为 ()

A spring has a **spring constant** $k = 50 \text{ N/m}$ and an original length of 10 cm . If it is stretched to 14 cm , the magnitude of the pulling force is ().

- A. 6 N
- B. 4 N
- C. 2 N
- D. 8 N

4. 质量 $m = 5 \text{ kg}$ 的物体放在水平面上，动摩擦因数 $\mu = 0.2$ 。用 $F = 8 \text{ N}$ 的水平拉力拉动物体，摩擦力大小为 () (取 $g = 10 \text{ m/s}^2$)

An object of mass $m = 5 \text{ kg}$ is placed on a horizontal surface. The **coefficient of kinetic friction** is $\mu = 0.2$. A horizontal force $F = 8 \text{ N}$ is applied to pull the object. The magnitude of the frictional force is (). (Take $g = 10 \text{ m/s}^2$)

- A. 8 N
- B. 4 N
- C. 0 N
- D. 10 N

5. 三个力大小分别为 3 N 、 4 N 、 5 N ，作用于同一物体，它们的合力最大值为 ()

Three forces with magnitudes of 3 N , 4 N , and 5 N act on the same object. The maximum magnitude of their **resultant force** is ().

- A. 6 N
- B. 12 N
- C. 10 N
- D. 0 N

6. 大小为 15 N 和 5 N 的两个力方向相反, 它们的合力大小为 ()

Two forces of magnitudes 15 N and 5 N act in **opposite directions**. The magnitude of their resultant force is ().

- A. 15 N
- B. 10 N
- C. 5 N
- D. 20 N

7. 力的三要素是大小、方向和 ()

The **three elements of a force** are magnitude, direction, and ().

- A. 性质 Nature
- B. 作用点 Point of application
- C. 受力对象 Object of action
- D. 作用效果 Effect

8. 物体由静止开始做匀加速直线运动, 加速度 $a = 3 \text{ m/s}^2$, 5 s 末的速度为 ()

An object starts from rest and moves with **uniformly accelerated linear motion** at $a = 3 \text{ m/s}^2$. Its velocity at the end of 5 s is ().

- A. 10 m/s
- B. 5 m/s
- C. 20 m/s
- D. 15 m/s

9. 质量 2 kg 的物体在水平面上受 10 N 的拉力, 动摩擦因数 0.2, 物体的加速度为 () (取 $g = 10 \text{ m/s}^2$)

An object of mass 2 kg is acted upon by a 10 N pulling force on a horizontal surface. If the coefficient of kinetic friction is 0.2, the acceleration of the object is (). (Take $g = 10 \text{ m/s}^2$)

- A. 6 m/s^2
- B. 3 m/s^2
- C. 4 m/s^2
- D. 5 m/s^2

10. 物体从倾角 30° 的光滑斜面顶端由静止滑下, 2 s 内的位移为 () (取 $g = 10 \text{ m/s}^2$)

An object slides down from rest from the top of a **smooth inclined plane** with an inclination angle of 30° . The displacement it travels in 2 s is (). (Take $g = 10 \text{ m/s}^2$)

- A. 10 m
- B. 20 m
- C. 5 m
- D. 15 m

11. 9 N 和 12 N 的两个力作用在 3 kg 的物体上, 物体的最大可能加速度为 ()

Two forces of 9 N and 12 N act on a 3 kg object. The maximum possible acceleration of the object is ().

- A. 4 m/s^2
- B. 5 m/s^2
- C. 6 m/s^2
- D. 7 m/s^2

12. 两个物体摩擦起电的过程中, 它们的总电荷量 ()

During the process of two objects being **charged by friction**, the total amount of charge they possess ().

- A. 保持不变 Remains constant
- B. 增加 Increases

- C. 条件不足无法确定 Insufficient conditions to determine
D. 减少 Decreases

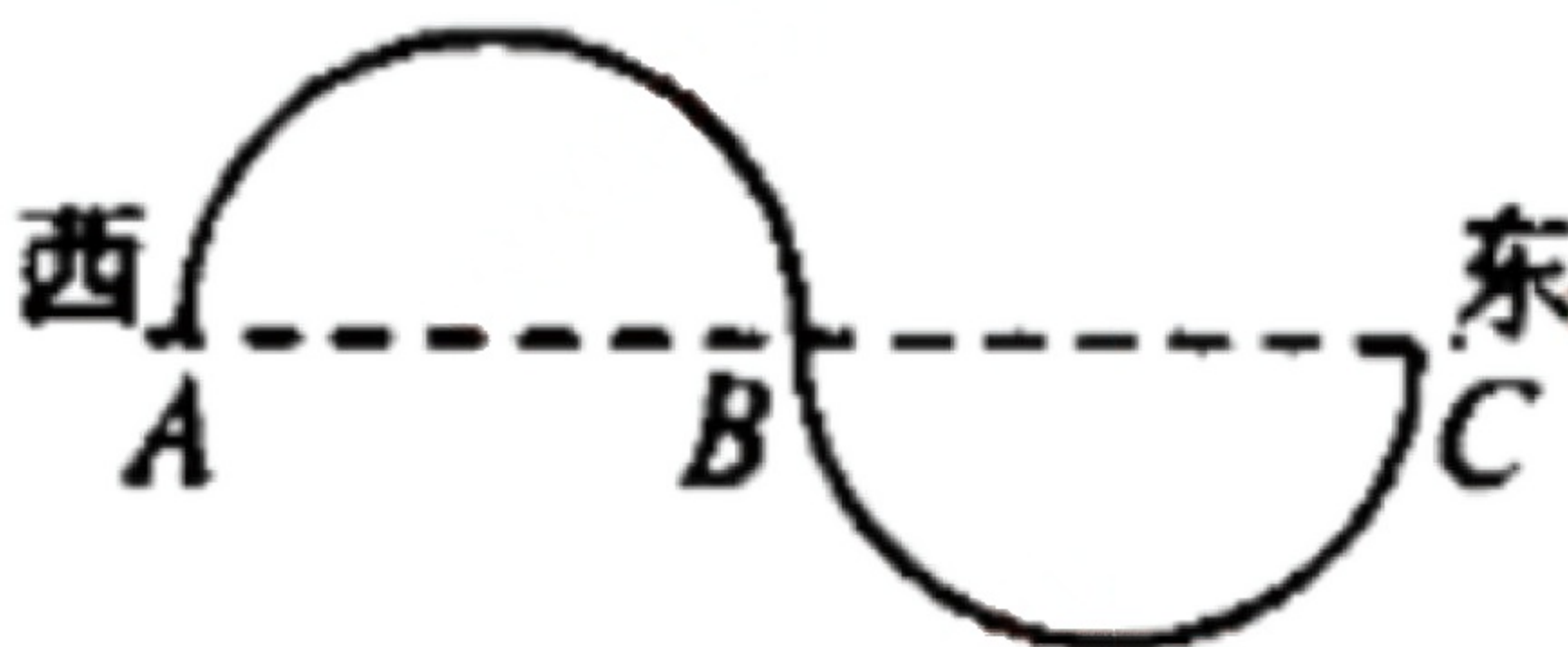
13. 质点做匀速圆周运动，线速度 $v = 4 \text{ m/s}$ ，半径 $R = 2 \text{ m}$ ，角速度为 ()

A particle undergoes **uniform circular motion** with a linear velocity of $v = 4 \text{ m/s}$ and a radius of $R = 2 \text{ m}$. Its **angular velocity** is ().

- A. 2 rad/s
B. 8 rad/s
C. 4 rad/s
D. 1 rad/s

14. 如图所示，物体沿半径 4 m 的半圆从 A 点运动到 C 点，用时 2 s ，平均速度大小为 ()

As shown in the figure, an object moves from point A to point C along a semicircle of radius 4 m , taking 2 s . The magnitude of its **average velocity** is ().



- A. 4 m/s
B. 0 m/s
C. 2 m/s
D. 8 m/s

15. 物体在水平面上做匀速直线运动，若所受摩擦力为 10 N ，拉力大小为 ()

An object moves with **uniform linear motion** on a horizontal surface. If the frictional force acting on it is 10 N , the magnitude of the pulling force is ().

- A. 0 N
B. 20 N
C. 10 N
D. 5 N

16. 弹簧受 10 N 拉力时长 12 cm ，受 20 N 拉力时长 14 cm ，弹簧原长为 ()

A spring has a length of 12 cm under a 10 N pulling force, and 14 cm under a 20 N pulling force. The original length of the spring is ().

- A. 8 cm
B. 11 cm
C. 9 cm
D. 10 cm

17. 下列关于力的说法错误的是 ()

Which of the following statements about force is **incorrect**? ()

- A. 力是物体运动状态改变的原因。Force is the cause of a change in an object's state of motion.
B. 力是矢量，既有大小又有方向。Force is a vector, having both magnitude and direction.
C. 力可以只有施力物体没有受力物体。A force can have an exerting object but no receiving object.
D. 力是物体对物体的作用。Force is an interaction between two objects.

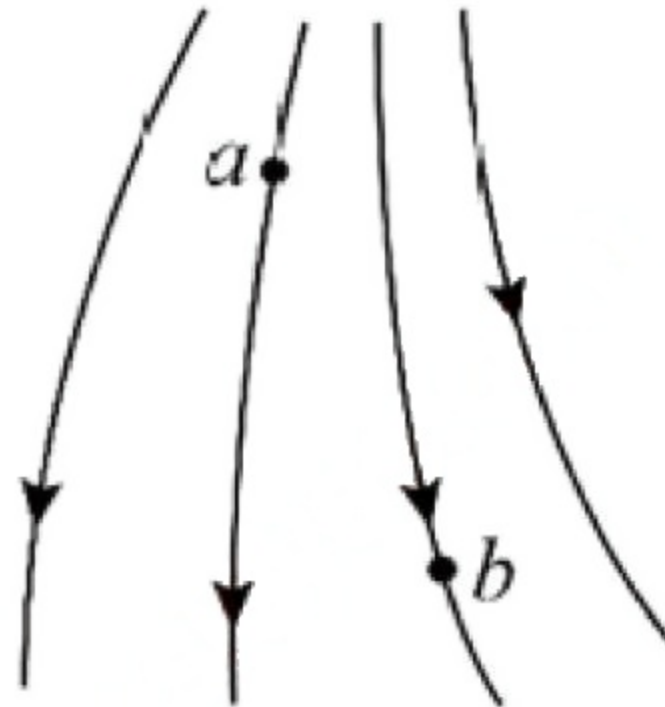
18. 质量 4 kg 的物体从倾角 30° 的斜面顶端由静止滑下, 3 s 内重力的冲量大小为 () (取 $g = 10 \text{ m/s}^2$)
An object of mass 4 kg slides down from rest from the top of an inclined plane with an inclination angle of 30° . The magnitude of the **impulse** due to gravity in 3 s is (). (Take $g = 10 \text{ m/s}^2$)
A. 60 N·s
B. 40 N·s
C. 80 N·s
D. 120 N·s
19. 15 N 的水平拉力使物体在粗糙水平面上移动 3 m, 若摩擦力为 5 N, 拉力做的功为 ()
A horizontal pulling force of 15 N moves an object a distance of 3 m on a rough surface. If the frictional force is 5 N, the **work done** by the pulling force is ().
A. 45 J
B. 60 J
C. 30 J
D. 15 J
20. 坐在匀速行驶的火车中, 窗外树木向后退, 所选参考系是 ()
When sitting in a train moving at a constant speed, the trees outside the window appear to move backward. The **reference frame** chosen is ().
A. 太阳 The Sun
B. 树木 The trees
C. 地面 The ground
D. 火车 The train
21. 弹簧振子完成一次全振动用时 0.5 s, 振动频率为 ()
A spring-mass system completes one full oscillation in 0.5 s. The **frequency** of oscillation is ().
A. 0.5 Hz
B. 1 Hz
C. 4 Hz
D. 2 Hz
22. 声波频率 $f = 400 \text{ Hz}$, 波速 $v = 340 \text{ m/s}$, 波长为 ()
A sound wave has a frequency of $f = 400 \text{ Hz}$ and a wave speed of $v = 340 \text{ m/s}$. Its **wavelength** is ().
A. 1.7 m
B. 3.4 m
C. 0.85 m
D. 0.425 m
23. 两个相同金属球分别带 +4 C 和 -6 C 的电荷, 接触后分开, 每个球的电荷量为 ()
Two identical metal spheres carry charges of +4 C and -6 C. After they are brought into contact and then separated, the charge on each sphere is ().
A. +2 C
B. -2 C
C. +1 C
D. -1 C
24. 物体质量 $m = 11 \text{ kg}$, 重力为 () (取 $g = 10 \text{ m/s}^2$)
An object has a mass of $m = 11 \text{ kg}$. Its weight is (). (Take $g = 10 \text{ m/s}^2$)
A. 110 N
B. 11 N

C. 0.11 N

D. 1.1 N

25. 电场线分布如图所示，a、b 两点放等量正电荷，电场力 F_a 与 F_b 的关系是 ()

The electric field lines are distributed as shown in the figure. Equal positive charges are placed at points a and b. The relationship between the electric forces F_a and F_b is ().



A. 无法确定 Cannot be determined

B. $F_a = F_b$

C. $F_a < F_b$

D. $F_a > F_b$

26. 100 Ω 和 200 Ω 的电阻并联，总电阻为 ()

Resistors of 100 Ω and 200 Ω are connected in **parallel**. The total resistance is ().

A. 150 Ω

B. 50 Ω

C. 300 Ω

D. 66.7 Ω

27. $R_1 = 6 \Omega$ 、 $R_2 = 4 \Omega$ 的两个电阻并联在 12 V 电源两端，电流之比 $I_1 : I_2$ 为 ()

Two resistors $R_1 = 6 \Omega$ and $R_2 = 4 \Omega$ are connected in parallel across a 12 V power supply. The ratio of their currents $I_1 : I_2$ is ().

A. 4:6

B. 2:3

C. 3:2

D. 1:1

28. 点电荷 Q 在 A 点产生的电场强度为 E，若点电荷到 A 点的距离变为原来的 2 倍，电场强度变为 ()

A point charge Q produces an **electric field strength** E at point A. If the distance from the point charge to point A doubles, the electric field strength becomes ().

A. $E/2$

B. $2E$

C. $E/4$

D. $4E$

29. 磁感应强度和磁通量的国际单位分别是 ()

The SI units of **magnetic flux density** and **magnetic flux** are respectively ().

A. 安培、库仑 Ampere, Coulomb

B. 特斯拉、韦伯 Tesla, Weber

C. 牛顿、焦耳 Newton, Joule

D. 伏特、欧姆 Volt, Ohm

30. 光从空气射入玻璃, 入射角 45° , 反射角为 ()

Light travels from air into glass. If the **angle of incidence** is 45° , the **angle of reflection** is ().

- A. 90°
- B. 60°
- C. 45°
- D. 30°

31. 如图所示, A、B 为吊扇上两点, 风扇匀速转动, 两点线速度 v 、角速度 ω 的关系正确的是 ()

As shown in the figure, A and B are two points on a ceiling fan. When the fan rotates at a constant speed, the correct relationship between the linear velocity v and angular velocity ω of the two points is ().



- A. $v_A < v_B$, $\omega_A < \omega_B$
- B. $v_A = v_B$, $\omega_A = \omega_B$
- C. $v_A < v_B$, $\omega_A = \omega_B$
- D. $v_A > v_B$, $\omega_A > \omega_B$

32. 长 0.5 m 的导线放在 $B = 0.4 \text{ T}$ 的磁场中, 所受最大安培力 0.2 N, 导线中电流大小为 ()

A wire of length 0.5 m is placed in a magnetic field of flux density $B = 0.4 \text{ T}$ and experiences a maximum **magnetic force** of 0.2 N. The magnitude of the current in the wire is ().

- A. 2 A
- B. 0.5 A
- C. 1 A
- D. 4 A

33. 一定质量的气体等容变化, 温度从 27°C 升至 54°C , 压强变为原来的 () 倍

A fixed amount of gas undergoes **isochoric change**. Its temperature increases from 27°C to 54°C . The pressure becomes () times its original value.

- A. 1.09
- B. 1
- C. 2
- D. 1.5

34. 50 匝的线圈, 磁通量在 0.1 s 内从 0.2 Wb 均匀增至 0.6 Wb, 感应电动势为 ()

A coil has 50 turns. The magnetic flux through it increases uniformly from 0.2 Wb to 0.6 Wb in 0.1 s. The **induced electromotive force (EMF)** is ().

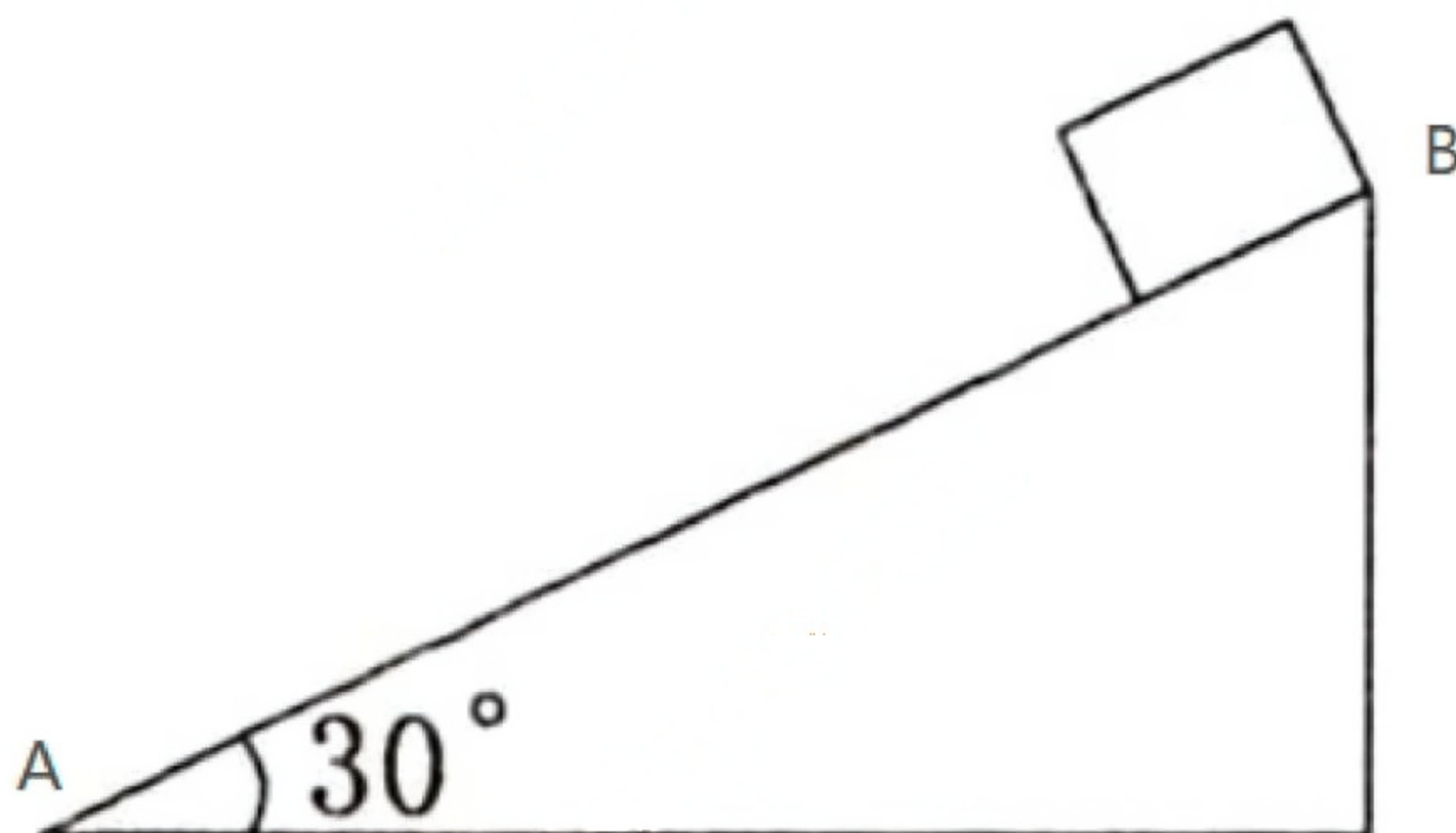
- A. 200 V
- B. 20 V
- C. 100 V
- D. 50 V

35. 质量 0.2 kg 的小球从 2 m 高处自由下落至地面, 重力势能变化量为 () (取 $g = 10 \text{ m/s}^2$)

A small ball of mass 0.2 kg falls freely from a height of 2 m to the ground. The change in its **gravitational potential energy** is (). (Take $g = 10 \text{ m/s}^2$)

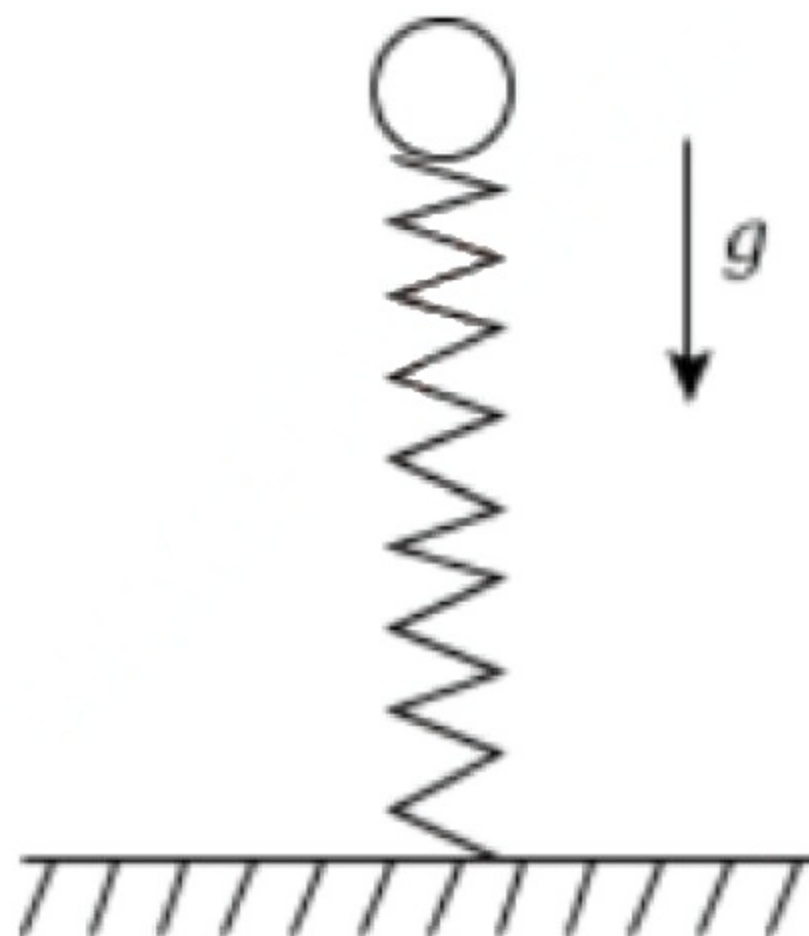
- A. 0 J
- B. -4 J

- C. 4 J
D. 2 J
36. 电动机电阻 $2\ \Omega$, 额定电压 100 V , 正常工作电流 3 A , 电动机发热功率损耗为 ()
The resistance of an electric motor is $2\ \Omega$, and the rated voltage is 100 V . When it operates normally, the current is 3 A . The **heating power loss** of the motor is ().
A. 300 W
B. 18 W
C. 200 W
D. 900 W
37. 光从光密介质射入光疏介质, 入射角大于临界角时, 发生 ()
When light travels from an **optically denser medium** to an **optically rarer medium** and the angle of incidence is greater than the **critical angle**, () occurs.
A. 全反射 Total internal reflection
B. 折射 Refraction
C. 反射 Reflection
D. 散射 Scattering
38. 真空中两个点电荷 $Q_A = +2 \times 10^{-9}\text{ C}$ 、 $Q_B = -2 \times 10^{-9}\text{ C}$, 相距 4 cm , 连线中点的电场强度大小为 () (静电力常量 $k = 9.0 \times 10^9\text{ N}\cdot\text{m}^2/\text{C}^2$)
Two point charges $Q_A = +2 \times 10^{-9}\text{ C}$ and $Q_B = -2 \times 10^{-9}\text{ C}$ in a vacuum are separated by 4 cm . The magnitude of the electric field strength at the midpoint of their connecting line is (). (Coulomb constant $k = 9.0 \times 10^9\text{ N}\cdot\text{m}^2/\text{C}^2$)
A. 0 N/C
39. 质点做简谐运动, 经过平衡位置时, 下列物理量最大的是 ()
For a particle undergoing **simple harmonic motion**, which of the following quantities reaches its maximum value when the particle passes through the **equilibrium position**? ()
A. 加速度 Acceleration
B. 回复力 Restoring force
C. 速度 Velocity
D. 位移 Displacement
40. 如图所示, 质量 5 kg 的物体从静止开始沿倾角 30° 的斜面从 B 点滑至 A 点, 下滑时摩擦力 5 N , BA 间距 2 m , 物体在 A 点的速度为 () ($g = 10\text{ m/s}^2$)
As shown in the figure, a 5 kg object, initially at rest, slides down a 30° incline from point B to point A. The frictional force is 5 N , and the distance BA is 2 m . The velocity of the object at point A is (). ($g = 10\text{ m/s}^2$)



- A. 4 m/s
- B. $2\sqrt{5}$ m/s
- C. 8 m/s
- D. 5 m/s

41. $+5 \times 10^{-6}$ C 的电荷从 A 点移到 B 点, 电场力做正功 1×10^{-3} J, 电势差 U_{AB} 为 ()
 A charge of $+5 \times 10^{-6}$ C is moved from point A to point B, and the electric force does 1×10^{-3} J of positive work. The **potential difference** U_{AB} is ().
- A. -200 V
 - B. 200 V
 - C. -0.2 V
 - D. 0.2 V
42. 20 N 的水平力使物体移动 10 m, 摩擦力 5 N, 此过程中物体动能增加量为 ()
 An object is moved 10 m by a 20 N horizontal force. The frictional force is 5 N. The increase in its **kinetic energy** during this process is ().
- A. 50 J
 - B. 200 J
 - C. 100 J
 - D. 150 J
43. 子弹初速度 600 m/s, 穿透第一块木板后速度为 400 m/s, 再射入第二块相同木板, 子弹末速度为 ()
 A bullet with an initial speed of 600 m/s penetrates a wooden plank and emerges at 400 m/s. If it enters a second identical plank, the bullet's final speed will be ().
- A. 0
 - B. 100 m/s
 - C. 50 m/s
 - D. 200 m/s
44. 如图所示, 小球从空中自由下落, 与正下方轻质竖直弹簧接触直至速度为零, 关于小球运动说法正确的是 ()
 As shown in the figure, a small ball falls freely and contacts a lightweight vertical spring below until its velocity reaches zero. Which statement about the ball's motion is correct? ()



- A. 接触后小球做减速运动, 加速度增大、速度减小至零
 After contact, the ball decelerates, with increasing acceleration and decreasing velocity until zero.
- B. 接触后速度为零、加速度为零的位置是弹簧压缩量最大处
 After contact, the position with zero velocity and zero acceleration is where the spring is most compressed.
- C. 接触后小球先加速后减速, 速度先增大后减小至零

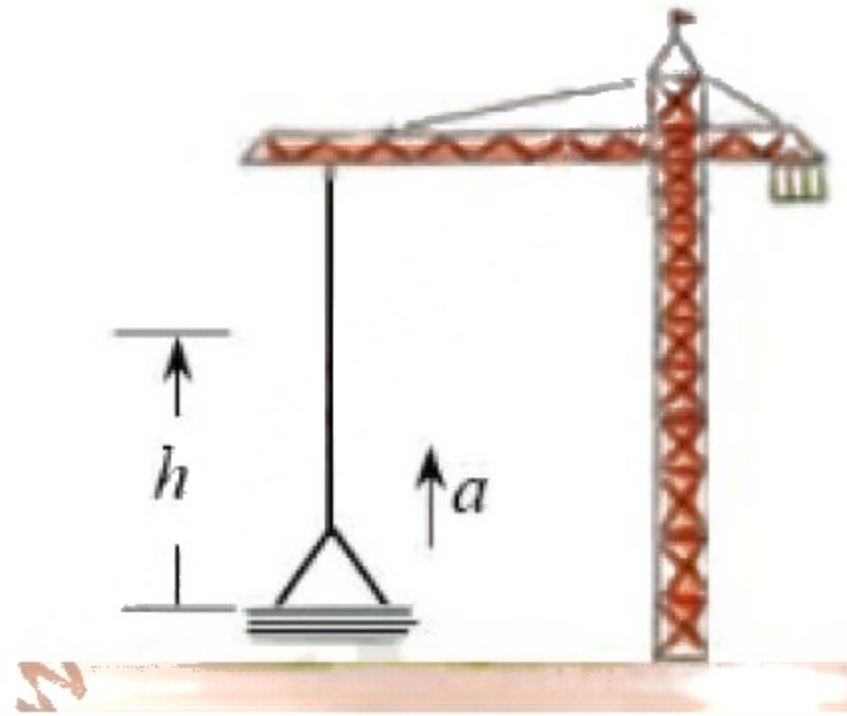
After contact, the ball first accelerates then decelerates; velocity increases first then decreases to zero.

D. 接触后小球速度最大的位置是弹簧压缩量最大处

The position of maximum velocity is where the spring is most compressed.

45. 如图所示，塔吊用钢缆将质量为 m 的建筑材料竖直向上提升高度 h ，重力加速度为 g ，关于该过程说法正确的是（ ）

As shown in the figure, a tower crane lifts a building material of mass m vertically upward by height h with a steel cable. The acceleration due to gravity is g . Which statement about the process is correct? ()



A. 钢缆拉力做功为 $m(a + g)h$

Work done by cable tension is $m(a + g)h$.

B. 重力对建材做功为 $-mah$

Work done by gravity is $-mah$.

C. 合外力对建材做功为 mgh

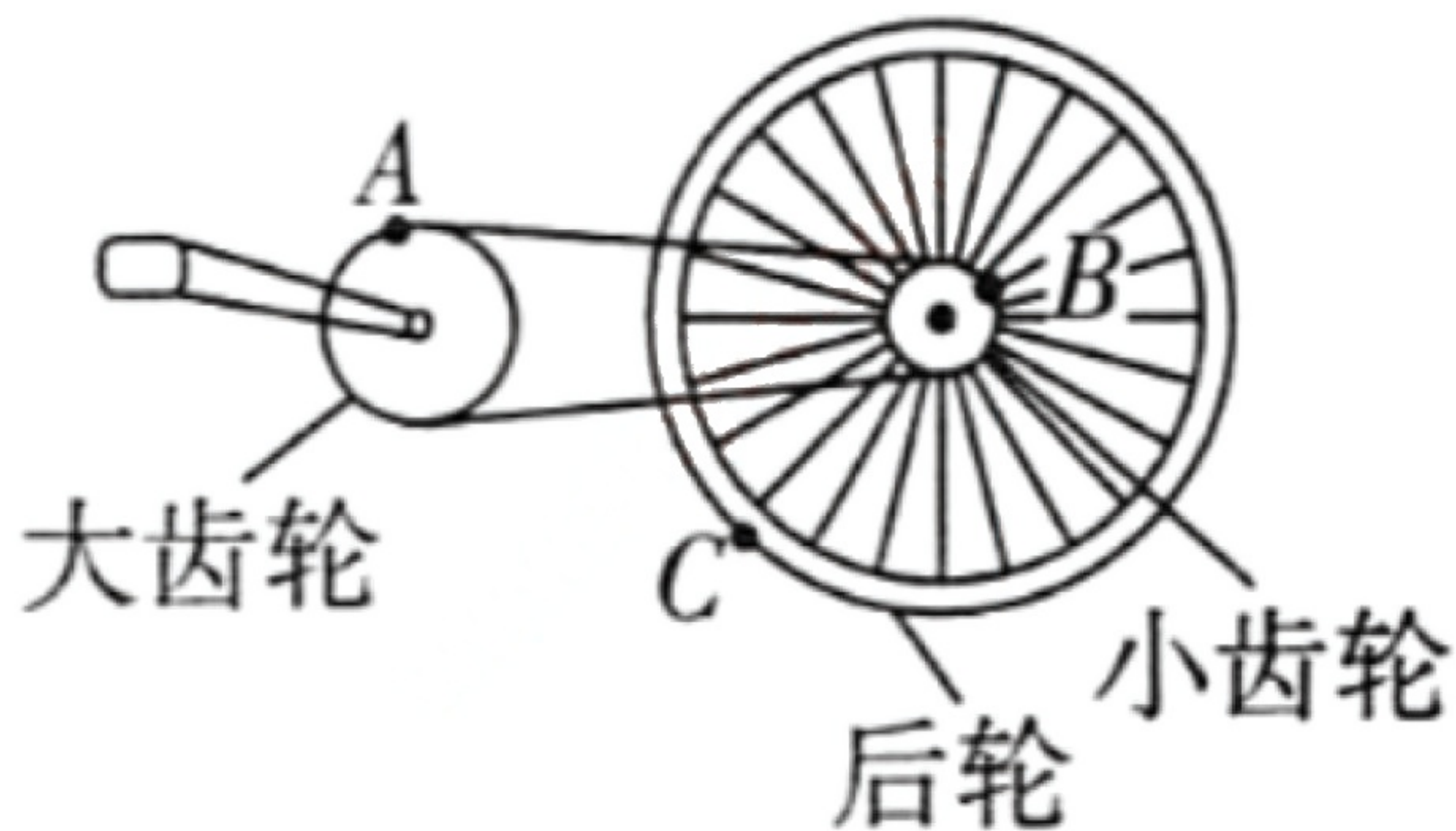
Work done by net force is mgh .

D. 建材重力势能减少 mgh

Gravitational potential energy decreases by mgh .

46. 如图所示，自行车有前牙盘、后飞轮、后轮，半径不同，边缘上 A、B、C 三点说法正确的是（ ）

As shown in the figure, a bicycle has a front chainwheel, rear sprocket, and rear wheel with different radii. Which statement about points A, B, C on their edges is correct? ()



A. B、C 线速度相同

Points B and C have the same linear speed.

B. A、B 线速度相同

Points A and B have the same linear speed.

C. A、C 线速度相同

Points A and C have the same linear speed.

D. A、B 角速度相同

Points A and B have the same angular velocity.

47. 汽车在平直公路上从初速度 v_0 开始加速，经时间 t 位移为 s ，达到最大速度 v_m 。此过程发动机功率恒定，阻力恒为 f ，发动机做功为（）

A car accelerates from initial speed v_0 on a straight road. After time t , displacement is s and it reaches maximum speed v_m . Engine power is constant, resistance is constant f . The work done by the engine is ().

A. fv_mt

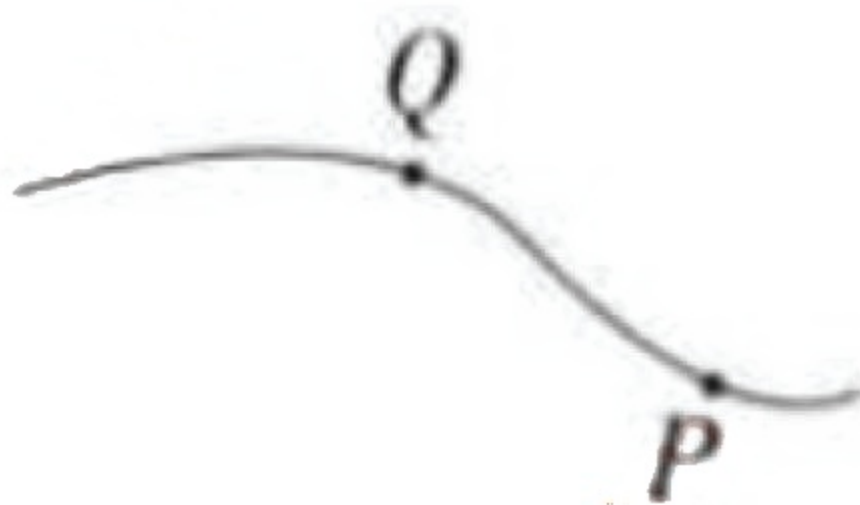
B. $\frac{1}{2}ft(v_0 + v_m)$

C. fs

D. $\frac{1}{2}mv_m^2 - \frac{1}{2}mv_0^2$

48. 风洞实验室用烟雾颗粒显示气流，某烟雾颗粒轨迹为曲线，说法正确的是（）

A wind tunnel uses smoke particles to visualize airflow. A smoke particle follows a curved trajectory. Which statement is correct? ()



A. 烟雾颗粒速度恒定

The particle's velocity is constant.

B. 烟雾颗粒一定做匀变速曲线运动

The particle must move with uniformly accelerated curvilinear motion.

C. P 点加速度方向可能水平向左

Acceleration at point P may be horizontally leftward.

D. Q 点合外力方向可能竖直向下

Net force at point Q may be vertically downward.

Answer

1

Answer: d

解析: 标量只有大小、无方向；矢量既有大小又有方向。位移、加速度、力是矢量，功是标量。

Explanation: A scalar has only magnitude; a vector has both magnitude and direction.

Displacement, acceleration, and force are vectors; **work is a scalar.**

2

Answer: b

解析: 地球重力 $G = mg = 6 \times 10 = 60 \text{ N}$ ，月球重力为地球的 $1/6$ ，即 $60 \times 1/6 = 10 \text{ N}$ 。

Explanation: Weight on Earth $G = mg = 60 \text{ N}$; weight on the Moon is $1/6$ of Earth's weight: 10 N .

3

Answer: c

解析: 胡克定律 $F = k\Delta x$, 形变量 $\Delta x = 14\text{ cm} - 10\text{ cm} = 0.04\text{ m}$, 拉力 $F = 50 \times 0.04 = 2\text{ N}$ 。

Explanation: By Hooke's Law $F = k\Delta x$. Extension $\Delta x = 0.04\text{ m}$, so $F = 2\text{ N}$.

4

Answer: a

解析: 最大静摩擦力 $\approx$ 滑动摩擦力 $f_{\max} = \mu mg = 10\text{ N}$; 拉力 $F = 8\text{ N} < 10\text{ N}$, 物体静止, 静摩擦力等于拉力, 为 8 N 。

Explanation: Maximum static friction $f_{\max} = 10\text{ N}$. Applied force $8\text{ N} < 10\text{ N}$, so the object is stationary; static friction equals applied force: 8 N .

5

Answer: b

解析: 三个力同向时合力最大, $3 + 4 + 5 = 12\text{ N}$ 。

Explanation: The resultant force is maximized when all forces act in the same direction: 12 N .

6

Answer: b

解析: 方向相反的两个力, 合力为大小之差, $15 - 5 = 10\text{ N}$ 。

Explanation: For forces in opposite directions, resultant force = difference of magnitudes: 10 N .

7

Answer: b

解析: 力的三要素: 大小、方向、作用点。

Explanation: The three elements of a force are magnitude, direction, and **point of application**.

8

Answer: d

解析: 初速度为 0 的匀加速直线运动, $v = at = 3 \times 5 = 15\text{ m/s}$ 。

Explanation: Uniform acceleration from rest: $v = at = 15\text{ m/s}$.

9

Answer: 3 m/s^2 (题目选项排版错误)

解析: 合力 $F_{\text{合}} = 10 - \mu mg = 6\text{ N}$, 加速度 $a = F_{\text{合}}/m = 3\text{ m/s}^2$ 。

Explanation: Net force $F_{\text{net}} = 6\text{ N}$, acceleration $a = 3\text{ m/s}^2$.

10

Answer: a

解析: 光滑斜面加速度 $a = g \sin 30^\circ = 5\text{ m/s}^2$, 位移 $x = \frac{1}{2}at^2 = 10\text{ m}$ 。

Explanation: On a smooth incline, $a = 5\text{ m/s}^2$, displacement $x = 10\text{ m}$.

11

Answer: 7 m/s^2 (题目选项排版错误)

解析: 两力同向时合力最大 $9 + 12 = 21 \text{ N}$, 加速度 $a = 21/3 = 7 \text{ m/s}^2$ 。

Explanation: Maximum net force (same direction) = 21 N , acceleration $a = 7 \text{ m/s}^2$.

12

Answer: a

解析: 摩擦起电遵循**电荷守恒定律**, 总电荷量保持不变。

Explanation: Charge is conserved during friction electrification; total charge remains constant.

13

Answer: a

解析: 匀速圆周运动角速度 $\omega = v/R = 4/2 = 2 \text{ rad/s}$ 。

Explanation: For uniform circular motion, angular velocity $\omega = v/R = 2 \text{ rad/s}$.

14

Answer: a

解析: 平均速度 = 位移 / 时间; 半圆位移为直径 8 m , 平均速度 $8/2 = 4 \text{ m/s}$ 。

Explanation: Average velocity = displacement / time. Displacement = diameter = 8 m , so average velocity = 4 m/s .

15

Answer: c

解析: 匀速直线运动受力平衡, 拉力与摩擦力大小相等, 为 10 N 。

Explanation: Force balance in uniform linear motion; tension equals friction: 10 N .

16

Answer: d

解析: 设原长为 L , 列方程 $10 = k(0.12 - L)$ 、 $20 = k(0.14 - L)$, 解得原长 $L = 10 \text{ cm}$ 。

Explanation: Solve the equations to find the original length $L = 10 \text{ cm}$.

17

Answer: c

解析: 力是物体间的相互作用, **必须同时有施力物体和受力物体**, 该选项说法错误。

Explanation: Force is an interaction between two objects; it cannot exist without a receiving object. This statement is incorrect.

18

Answer: d

解析: 重力的冲量 $I = mgt = 4 \times 10 \times 3 = 120 \text{ N}\cdot\text{s}$ 。

Explanation: Impulse of gravity $I = mgt = 120 \text{ N}\cdot\text{s}$.

19

Answer: a

解析: 拉力做功 $W = Fs = 15 \times 3 = 45 \text{ J}$, 与摩擦力无关。

Explanation: Work done by tension $W = Fs = 45 \text{ J}$, independent of friction.

20

Answer: d

解析: 以火车为参考系, 窗外树木相对火车向后运动。

Explanation: Taking the train as the reference frame, trees move backward relative to the train.

21

Answer: d

解析: 频率 $f = 1/T = 1/0.5 = 2 \text{ Hz}$ 。

Explanation: Frequency $f = 1/T = 2 \text{ Hz}$.

22

Answer: c

解析: 波长 $\lambda = v/f = 340/400 = 0.85 \text{ m}$ 。

Explanation: Wavelength $\lambda = v/f = 0.85 \text{ m}$.

23

Answer: d

解析: 相同金属球接触后, 电荷先中和再平分, $(4 - 6) \div 2 = -1 \text{ C}$ 。

Explanation: Identical metal spheres neutralize and share charge equally: -1 C per sphere.

24

Answer: a

解析: 重力 $G = mg = 11 \times 10 = 110 \text{ N}$ 。

Explanation: Weight $G = mg = 110 \text{ N}$.

25

Answer: d

解析: 电场线越密, 场强越大; a 点场强大于 b 点, 等量正电荷的电场力 $F_a > F_b$ 。

Explanation: Denser field lines mean stronger electric field. $E_a > E_b$, so $F_a > F_b$ for equal positive charges.

26

Answer: d

解析: 并联电阻公式 $1/R = 1/100 + 1/200$, 解得 $R \approx 66.7 \Omega$ 。

Explanation: Parallel resistance formula gives $R \approx 66.7 \Omega$.

27

Answer: b

解析: 并联电路电压相等, 电流与电阻成反比, $I_1 : I_2 = R_2 : R_1 = 2 : 3$ 。

Explanation: In parallel, current is inversely proportional to resistance: $I_1 : I_2 = 2 : 3$.

28

Answer: c

解析: 点电荷场强 $E \propto 1/r^2$; 距离加倍, 场强变为原来的1/4。

Explanation: Point-charge field $E \propto 1/r^2$; doubling the distance reduces E to 1/4 of original.

29

Answer: b

解析: 磁感应强度的国际单位是**特斯拉**, 磁通量的国际单位是**韦伯**。

Explanation: SI units: magnetic flux density (Tesla), magnetic flux (Weber).

30

Answer: c

解析: 光的反射定律: 反射角等于入射角, 为 45° 。

Explanation: Law of reflection: angle of reflection = angle of incidence = 45° .

31

Answer: c

解析: 同轴转动角速度相等 $\omega_A = \omega_B$; 线速度 $v = \omega r$, B 半径更大, 故 $v_A < v_B$ 。

Explanation: Coaxial rotation: $\omega_A = \omega_B$. Larger radius at B means $v_A < v_B$.

32

Answer: c

解析: 最大安培力 $F = BIL$, 电流 $I = 0.2/(0.4 \times 0.5) = 1$ A。

Explanation: Maximum ampere force $F = BIL$, so current $I = 1$ A.

33

Answer: a

解析: 等容变化 $P \propto T$ (热力学温度); $T_1 = 300$ K, $T_2 = 327$ K, $P_2/P_1 = 1.09$ 。

Explanation: Isochoric process: $P \propto T$ (Kelvin). $P_2/P_1 = 1.09$.

34

Answer: a

解析: 法拉第电磁感应定律 $E = n\Delta\Phi/\Delta t = 50 \times (0.6 - 0.2)/0.1 = 200$ V。

Explanation: Faraday's Law: $E = 200$ V.

35

Answer: b

解析: 重力势能变化量 $\Delta E_p = -mgh = -0.2 \times 10 \times 2 = -4$ J。

Explanation: Change in gravitational potential energy $\Delta E_p = -4$ J.

36

Answer: b

解析: 电动机发热功率 $P = I^2 R = 3^2 \times 2 = 18 \text{ W}$ 。

Explanation: Heating power of the motor $P = I^2 R = 18 \text{ W}$.

37

Answer: a

解析: 光从光密介质射入光疏介质，入射角大于临界角时，发生**全反射**。

Explanation: Total internal reflection occurs under these conditions.

38

Answer: $9 \times 10^4 \text{ N/C}$ (题目选项缺失)

解析: 中点场强为两电荷场强之和，计算得 $9 \times 10^4 \text{ N/C}$ 。

Explanation: Field at midpoint = sum of fields from both charges: $9 \times 10^4 \text{ N/C}$.

39

Answer: c

解析: 简谐运动经过平衡位置时，**速度最大**，位移、回复力、加速度均为 0。

Explanation: At the equilibrium position of SHM, velocity is maximum; others are zero.

40

Answer: a

解析: 由动能定理 $mgx \sin 30^\circ - fx = \frac{1}{2}mv^2$ ，代入数据解得 $v = 4 \text{ m/s}$ 。

Explanation: Work-energy theorem gives $v = 4 \text{ m/s}$.

41

Answer: b

解析: 电势差 $U_{AB} = W/q = 1 \times 10^{-3} / 5 \times 10^{-6} = 200 \text{ V}$ 。

Explanation: Potential difference $U_{AB} = 200 \text{ V}$.

42

Answer: d

解析: 动能增加量等于合外力做功， $\Delta E_k = (20 - 5) \times 10 = 150 \text{ J}$ 。

Explanation: Increase in kinetic energy = work done by net force = 150 J.

43

Answer: a

解析: 穿透一块木板的动能损耗固定，第二块木板耗尽子弹剩余动能，末速度为 0。

Explanation: The second plank exhausts all kinetic energy; final speed = 0.

44

Answer: c

解析: 小球接触弹簧后，先重力大于弹力**加速**，后弹力大于重力**减速**，速度先增后减至 0。

Explanation: The ball accelerates first ($mg > kx$) then decelerates ($kx > mg$); velocity rises then falls to zero.

45

Answer: a

解析: 建材加速上升时, 拉力 $F = m(g + a)$, 拉力做功 $W = Fh = m(g + a)h$ 。

Explanation: Tension $F = m(g + a)$, work done by tension $W = m(g + a)h$.

46

Answer: b

解析: 自行车链条传动, 前牙盘 A 和后飞轮 B 边缘 **线速度大小相等**。

Explanation: Chain drive: points A and B have equal linear speed.

47

Answer: a

解析: 恒定功率 $P = fv_m$, 发动机做功 $W = Pt = fv_mt$ 。

Explanation: Constant power $P = fv_m$, work done by engine $W = fv_mt$.

48

Answer: d

解析: 曲线运动合外力指向轨迹凹侧, Q 点合外力方向 **可以竖直向下**。

Explanation: Net force points to the concave side; a downward net force is valid at Q.